Formulas for success

So far we've covered how to
start a new spreadsheet, enter in data, and make it look refined and ready for
some serious analysis. Now we'll learn how to perform calculations in your spreadsheet. You may need to
calculate everything from sums to averages, to finding minimum
and maximum amounts. You'll use calculations for a lot of different
kinds of tasks. In this video, we'll focus
on learning the basics and then do a little math with some sales data to practice. Let's talk about formulas first. You might remember that
a formula is a set of instructions that perform
a specific calculation. Basically, formulas can
do the math for you. Now, they don't only do math, they can do a lot more. Soon you'll learn different ways you can use them throughout the data
analysis processes. Formulas are built on operators
which are symbols that name the type of operation or calculation to be performed. For example, a plus sign
is a common operator. The formulas you use
as a data analyst will usually include at
least one operator. Now, let's talk about math
expressions or equations. These can take a lot
of different forms, but you might be familiar
with them already. 3 minus 1, 15 plus 8 divided
by 2, 846 times 513. These are all examples
of expressions. Is this bringing back
memories of grade school? Well, back in math class, you most likely learned to
complete an expression by including an equal
sign and the solution. It's slightly different
with spreadsheets. When you create a formula using an expression in a spreadsheet, you start the formula
with an equal sign. For example, if we
want to subtract, we type an equal sign
followed by the rest of the expression without any
spaces in the formula. Now let's try an expression that's a bit more challenging. We'll type 31982, then a hyphen for a minus
sign, then 17795. To calculate, we press "Enter." You'll most likely use formulas this way
when dealing with large numbers or expressions
with multiple steps. Here are the operators you

will use to complete formulas. The plus sign for addition, the minus or hyphen
for subtraction, the asterisk for multiplication, and the forward
slash for division. The division and
multiplication symbols might be different than
what you're used to. Small changes, but
important to keep in mind. If you already have data
in your spreadsheet, you can use cell references
in your formulas instead. A cell reference is
a single cell or range of cells in a worksheet that can
be used in a formula. Cell references
contain the letter of the column and the number of
the row where the data is. A range of cells is a
collection of two or more cells. A range can include cells
from the same row or column, or from different columns
and rows collected together. We'll show you an example
in an upcoming video. Now let's apply what we just
learned to some sales data. If we want to add
these figures to find the total sales for
the first row of data, you can click "cell F2". From there, we'll start
with an equal sign and use the cell references to input
values in your expression. We're starting with cell
B2 because the year in A2 is not a value we want
to add to the total. Then press "Enter." Just like that, your total sales has
been calculated for you, but what if you realized one of the values in your
data was wrong? No problem. You can change the
value in any cell using the formula and the total
will update automatically. The great thing about using
cell references is that they also automatically update when a formula is copied
to a new cell. Talk about a time-saver. Instead of entering
the same formula again for every new set
of cell references, just copy the formula using the menu or a keyboard
shortcut like Control plus C. Then paste the formula where
you want to apply it using Control plus V. And presto! The formula updates all the new cells and
values correctly. Now let's say you also want
it to find the average sales. For this, you create a new
formula in a different cell. To group values in a
formula, use parentheses. This lets your spreadsheet know
which values to calculate together and the order of the
operations to be performed. For example, open parentheses, then B2 plus C2 plus D2 plus E2,
and close parentheses,
then divide the value of all of this
by typing slash four. You are adding the values
in the four cells together and then using the slash to
divide the total by four, and just like the last one, we can copy and

paste the formula. Here's another formula you
can use if you want to find the percent change in sales
between June and July. Once a formula
calculates the value, you can then use the percent button to change the value
to a percentage. When you apply the formula
to the other rows, both the formula and the percent will
automatically update. That doesn't look like
the right answer. Looks like we've got
an error. Don't worry. Errors can happen at any
stage of data analysis, and that includes when
you're using spreadsheets. A formula has to be air tight. If there's something
wrong with one of the cell references,
it won't work. So what's our error? Well, we can see that the
value in cell D4 is missing. It might take some
time and research on your part to find the correct
value, but it's worth it. You want your analysis to
be as accurate as possible. When you do add the value, the formula takes
care of the rest. That was a lot to take in. Thanks for staying with me. You'll be able to apply
what you learned about formulas
here and later in the program to make your analysis more
efficient and your job, a little easier, and soon you'll work in
your own spreadsheet. Happy spreadsheeting.